

Backtracking Notes & Problem Roadmap

Goal: Master backtracking from fundamentals to interview-level optimizations.

Foundation

- Understand template: State, Choices, Constraints, Base Case, Undo.
- Template: Choose → Explore → Unchoose.

Phase 1: Basic Generation

- Subsets, Subsets II
- Combinations, Combination Sum I/II/III
- Permutations I/II

Phase 2: Strings

- Letter Combinations
- Generate Parentheses
- Restore IP Addresses
- Palindrome Partitioning

Phase 3: Grid

- Word Search
- Rat in a Maze
- Unique Paths III
- Path with Maximum Gold

Phase 4: Constraint Satisfaction

- N Queens: Normal → Boolean arrays → Bitmask
- Sudoku: Naive → Row/Col/Box sets → Bitmask

Phase 5: Partition

- Partition to K Equal Sum Subsets: Backtracking → Pruning → Bitmask+Memo
- Matchsticks to Square: Backtracking → Pruning → Bitmask+Memo

Phase 6: Graph

- All Paths Source to Target
- Hamiltonian Path
- Graph Coloring

Phase 7: Branch & Bound

- Job Assignment
- Travelling Salesman

Phase 8: Memoization

- Can I Win
- Repeated-state DFS + Memoization

Phase 9: Bitmask

- Beautiful Arrangement
- Advanced N Queens
- Advanced Sudoku

Optimisation Roadmap

Problem	V1	V2	V3
N Queens	Normal	Boolean arrays	Bitmask
Sudoku	Naive	Row/Col/Box sets	Bitmask
Combination Sum	Basic	Pruning	-
Word Search	Visited array	In-place marking	Trie + Backtracking
Partition K	Basic	Pruning	Bitmask + Memo
Matchsticks	Basic	Pruning	Bitmask + Memo

Recommended Solving Order

1. Template
2. Subsets
3. Subsets II
4. Combinations
5. Combination Sum I
6. Combination Sum II
7. Combination Sum III
8. Permutations
9. Permutations II
10. Letter Combinations
11. Generate Parentheses
12. Restore IP Addresses

13. Palindrome Partitioning
14. Word Search
15. Rat in a Maze
16. Path with Maximum Gold
17. Unique Paths III
18. N Queens
19. Sudoku Solver
20. Partition to K Equal Sum Subsets
21. Matchsticks to Square
22. All Paths Source to Target
23. Hamiltonian Path
24. Graph Coloring
25. Branch & Bound